

Ayush Kumar Singh

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Hostel-13, BIT Mesra, Ranchi (India) - 835215

EDUCATION

• Birla Institute of Technology, Mesra

Sept 2023 – Sept 2028

Integrated M.Sc. in Physics (Minor in Mathematics & Computing)

Ranchi, India

◦ CGPA: 8.72 / 10.0 (Branch Topper)

◦ 4th Semester SGPA: 9.44

Relevant Coursework: Classical Mechanics, Quantum Mechanics, Mathematical Physics, Computational Physics, Statistical Physics, Thermal Physics, Waves and Optics, General Relativity.

PUBLICATIONS & CONFERENCE PRESENTATIONS

• *Journal of Astrophysics and Astronomy* (Springer)

3rd author reviewing draft

Manuscript in preparation

◦ **Title:** UVIT Study of the Magellanic Clouds (U-SMAC) V. UV Stellar Populations in the Western Outskirts of the Small Magellanic Cloud

◦ **Authors:** Ayush Singh, [Sipra Hota](#), [Prasanta Kumar Nayak](#), and [Annapurni Subramaniam](#)

• A Decade of AstroSat Observations: Science Outcomes and Future Prospects

January 2026

Host Organisation: [U R Rao Satellite Centre \(URSC\)](#), Bengaluru, India



◦ Presented a **poster** titled “Exploring the Western Outskirts of the Small Magellanic Cloud: Tracing Foreground and SMC Stars with UVIT”, co-authored with [Sipra Hota](#), [Prasanta Kumar Nayak](#), and [Annapurni Subramaniam](#).

RESEARCH EXPERIENCE

• Birla Institute of Technology, Mesra, India

From April 2026

Title: Introduction to Bayesian Methods and Monte Carlo Techniques

On-site

Advisor: [Dr. Prakash Chandra](#)

◦ Studying fundamentals of Bayesian methods and implementing basic probabilistic models for parameter estimation.

◦ Exploring Monte Carlo methods, focusing on sampling techniques as part of ongoing research training.

• Indian Institute of Astrophysics, India

Ongoing from May 2025

Title: Disentangling Stellar Populations in the Western Halo of the SMC Using UVIT and Gaia DR3

Hybrid

Advisors: [Sipra Hota](#), [Dr. Prasanta Kumar Nayak](#), and [Prof. Annapurni Subramaniam](#)

◦ Conducted literature review and processed UVIT and Gaia DR3 data, including photometric calibration and generation of clean catalogs.

◦ Applied advanced analysis techniques such as cross-matching, vector point diagrams, GMM, and KDE to study stellar kinematics and separate Milky Way foreground (including 47 Tuc) from SMC members.

◦ Constructed UV/optical CMDs and color-color diagrams, performed isochrone fitting, and identified distinct stellar populations. *Manuscript in preparation*

• Birla Institute of Technology, Mesra, India

3 months: April 2025 – June 2025

Title: Quasi-Normal Modes of Ellis–Bronnikov Wormholes

On-site

Advisor: [Dr. Suman Ghosh](#)

◦ Investigated quasi-normal modes of Ellis–Bronnikov wormholes under perturbations.

◦ Implemented and compared WKB and Prony methods for QNM computation.

• Birla Institute of Technology, Mesra, India

1 year: May 2024 – April 2025

Title: Performance Analysis and Optimization of a Consumption-Scale Biogas Plant

On-site

Advisor: [Dr. Rajeev Kumar](#)

- Led the setup, data collection, and maintenance of a newly built biogas plant.
- Analyzed system data to optimize efficiency, waste ratios, and mechanical performance.
- Assisted Bachelor, Master, and Ph.D. students in data analysis and thesis preparation.
- Developed Project that won an [INR 10,000](#) award at the BIT NISHAN Business Competition.

WORKSHOPS AND SUMMER COURSES

• KSO Summer School, Indian Institute of Astrophysics

10 days: June 2025

Kodaikanal, India

On-site

- Participated in hands-on sessions on astronomy, including data cleaning, and basic analysis techniques.
- Attended lectures on different branches of astronomy including general relativity, planetary science, solar studies, stellar evolution, and active galactic nuclei .

PROGRAMMING & COMPUTATIONAL SKILLS

- **Programming Languages & Tools:** Python, R (statistics), MATLAB, C, C++, $\text{\LaTeX}$, Linux Shell Scripting
- **Scientific Computing (Python):** NumPy, SciPy, Pandas, AstroPy, Astroquery, Numerical Techniques
- **Data Visualization:** Matplotlib
- **Astronomical Data Analysis:** SAO DS9, Aladin Sky Atlas, TOPCAT, IRAF (photometry), DAOPHOT (PSF fitting), CCDLAB (FITS reduction), GLUE
- **Catalogs & Survey Data:** Gaia DR3, Astrosat/UVIT
- **Modeling & Simulations:** Stellar evolution (PARSEC)

HONORS AND AWARDS

• Branch Topper Award, Birla Institute of Technology, Mesra, India

Coordinator: [Dr. Rajeev Kumar](#)

- Received a cash prize of INR 15,000 for securing the highest academic rank in the Physics branch during the 2nd–4th semesters, recognizing consistent academic excellence and top departmental performance.

VOLUNTEER EXPERIENCE

• Volunteer for Rural Development, [Unnat Bharat Abhiyan \(UBA\)](#)

August 2024 – Present

Coordinator: [Dr. Rajeev Kumar](#)

- Contributing to rural development through Unnat Bharat Abhiyan by engaging in projects aimed at improving infrastructure, sanitation, and education access in villages, while enhancing leadership, problem-solving, and community engagement skills.

ADDITIONAL INFORMATION

Languages:

- Hindi (Native)
- Maithili (Native)
- English (C1)
- German (Beginner)

Interests:

- Wildlife
- Philosophy
- Politics
- Debate